

DermAI

Project Proposal



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A handwritten signature in blue ink, appearing to read 'Taha', is placed next to the name.

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1. Introduction

Regardless of age, location, or socioeconomic status, skin disorders impact millions of people worldwide, making it one of the most pervasive health issues. Timely diagnosis of skin problems is a significant challenge in areas with limited access to dermatologists, such as many rural and semi-urban areas. Even common illnesses like eczema or fungal infections can worsen into more serious health problems if they are not addressed or are misdiagnosed. As desktop and mobile technology become more widely available, artificial intelligence (AI) has become a promising tool for facilitating early, accessible, and reasonably priced healthcare diagnostics.

Designing and creating an AI-powered skin disease diagnosis system that can be used as a desktop and mobile application is the goal of this project. The system will categorize user-submitted skin photos into two groups: eczema and fungal infection using a trained convolutional neural network (CNN). These situations were picked because open-source labeled datasets are available, they are highly prevalent, and they have distinct visual features. The program will include a cross-questioning feature driven by a Large Language Model (LLM) to improve diagnosis accuracy and get around the drawbacks of image-only solutions. In order to enable the model to deliver a more comprehensive and contextually aware diagnosis, the system will pose pertinent follow-up questions following an initial prediction (such as sensations like itching, soreness, or swelling).

In its early stages, the system will only support English and run online. Through the integration of natural language interaction with image classification, the suggested approach offers a hybrid diagnostic experience that is both user-focused and technically sophisticated. In order to help users better understand their skin concerns and to encourage prompt consultation with a healthcare expert, the program is intended to act as a preliminary diagnostic helper. This method not only shows how computer vision and natural language processing (NLP) may be used practically, but it also advances the larger objective of improving the intelligence, accessibility, and interactivity of AI-driven healthcare.

2. Project Scope

The design and development of an AI-powered skin disease diagnosis application that employs interactive symptom-based inquiry and image-based analysis to classify 5-7 distinct conditions— like, fungal infection and eczema—is the main goal of this final year project. The web application, which will be compatible with desktop and mobile platforms, will enable users to get initial diagnostic advice and, if necessary, seek medical advice. The diagnosis of skin conditions outside than the two chosen categories will not be covered by the initiative. It won't enable high-resolution dermatoscopic pictures commonly used in clinical settings or provide real-time video-based diagnostics. Furthermore, the system will function as a helpful, pre-diagnostic tool rather than a direct medical prescription or a substitute for a professional diagnosis.

A predetermined set of medically pertinent questions specific to the two supported illnesses will be implemented as part of the project's LLM-powered cross-questioning capability. Building a general-purpose medical chatbot that can manage a variety of illnesses or symptoms outside of skin health is not its goal. We anticipate that people will use devices with rudimentary cameras, have a reliable internet connection, and provide crisp pictures of the afflicted skin area. The doctor consultation feature is predicated on the presence of certified medical personnel who are either on call or integrated with an outside service. In its first iteration, the system will only handle English; multilingual support is deemed outside the purview of this project.

3. Objective

1. This project's main goal is to create a cross-platform, AI-powered application for desktop and mobile devices that uses convolutional neural networks (CNNs) to classify images in order to identify **5 to 7 common skin conditions** — including infections, inflammatory issues, allergic reactions, and chronic dermatological disorders.
2. For immediate, automated analysis, users will be able to upload or capture images of affected skin areas.
3. The application will include a large language model (LLM)-powered intelligent cross-questioning module, which will ask follow-up questions based on symptoms (e.g., itching, redness, scaling) to enhance diagnostic accuracy.

4. By combining both **visual and contextual information**, the system aims to deliver more reliable results.
5. The application will support English-language interaction and run fully online.
6. The ultimate objective is to provide a quick, interactive, and scalable AI solution for the **early detection of 5–7 major skin conditions**, helping users make informed decisions about seeking professional care.

4. Problem Description

Common skin conditions such as eczema and fungal infections are often misdiagnosed or left untreated, especially in regions with limited access to dermatologists. Traditional diagnosis requires in-person consultations, which can be expensive, time-consuming, and inaccessible for people living in rural or underdeveloped areas. On the other hand, self-diagnosis is highly unreliable and can lead to worsening conditions or ineffective treatments. This highlights a critical gap in dermatological care, creating an urgent need for accessible, intelligent diagnostic tools. To address this, we propose an AI-powered application that allows users to identify skin diseases by uploading a picture of the affected area. However, image-only predictions may not always be accurate, particularly when different conditions present similar visual symptoms. To overcome this, our solution integrates a Large Language Model (LLM)-based cross-questioning module, which interacts with users through relevant follow-up questions such as “Is the area itchy?” or “Do you have a history of allergies?” to collect contextual information that enhances diagnostic precision. This hybrid approach not only improves accuracy but also builds user trust in AI systems by mimicking a human-like consultation experience. While developing this system, it is essential to consider challenges such as data privacy, model bias across skin tones, and the risks of false positives or negatives. Ultimately, the goal is to create a reliable, scalable, and user-friendly solution that empowers individuals with early-stage, trustworthy insights into their skin health—without replacing professional medical care, but instead supplementing it where access is limited.

5. Literature Review

Literature Review

The recent developments in artificial intelligence (AI), specifically in computer vision and natural language processing (NLP), have brought new avenues for the development of healthcare solutions targeting disease detection and patient interaction. AI application in dermatology has been significantly impacted by the availability of datasets like HAM10000 and ISIC containing thousands of labeled dermoscopic images for different skin ailments (Tschandl et al., 2018). These datasets are now a standard against which convolutional neural networks (CNNs) are trained and tested for skin lesion classification.

A very prominent paper in this research area was published by Esteva et al. (2017), in which a CNN was trained to diagnose skin cancer on more than 129,000 clinical images. The algorithm achieved the performance of dermatologists, indicating the potential of deep learning for medical image analysis. But these models are usually restricted to extreme or uncommon instances like melanoma and do not transfer well to frequent diseases like eczema or fungal infections, which are more common in underdiagnosed groups.

In line with this, additional research by Han et al. (2020) highlighted the significance of dataset diversity for model performance. In their research on shared skin ailments, they showed that CNNs had improved performance when trained on data that included a broad span of skin tones, lighting environments, and image resolutions. This has special implications for real-world applications to populations with diversity. Moreover, their findings also reinforced the idea that models concentrating on fewer, well-defined classes are better in confined settings such as mobile devices.

Concurrently, extensive research has been performed on light-weight CNN models that are best suited for mobile and edge deployment. Howard et al. (2019) presented MobileNetV2, which employed depthwise separable convolutions and inverted residual blocks to keep accuracy intact while cutting down computation. Tan and Le (2019) also presented EfficientNet, which scales width, depth, and resolution using compound scaling. Both architectures are appropriate for real-time systems and have found applications in mobile health diagnostics where size and speed are the primary constraints.

Despite these advances, image-based models are inherently limited. They are unable to capture unseen symptoms like itching, pain, or patient history. To address this limitation, researchers have begun to investigate multimodal methods. Wu et al. (2022) presented a medical chatbot that improves diagnostic ability through questioning users based on their symptoms. The authors blended NLP and machine learning to harvest context unavailable in image data and demonstrated better performance in differential diagnosis.

New NLP models such as BERT (Devlin et al., 2019) and GPT-3 (Brown et al., 2020) have revolutionized the way machines understand human language. BERT has been adapted for question-answering in clinical applications (Lee et al., 2020) with high accuracy in understanding pertinent information from patient responses. Likewise, clinical versions including BioBERT and MedGPT have proved useful in understanding health-related dialogues and hence can be utilized for cross-questioning modules of AI health systems.

Furthermore, the fusion of image and text data for enhanced diagnosis has been explored in publications such as that of Singh et al. (2021), where CNNs were combined with symptom-based survey inputs for diabetic retinopathy diagnosis. Their hybrid model registered higher accuracy when compared to image-only models. Nonetheless, the majority of these studies focus on internal diseases or specialist clinical settings and do not aim at common, external skin diseases that strike sizable portions of the population.

In summary, although both image-based classification and language-based diagnostic support have sizeable bodies of research, few systems integrate the two disciplines into one, interactive platform. This FYP fills that void by recommending a dual-platform (desktop and mobile) hybrid approach. It marries the capability of CNNs (for image classification) with LLM-generated cross-questioning (for contextual symptom analysis), targeted specifically at early-stage, non-critical skin conditions such as eczema, rashes, and fungal infections. Through this, it adds a scalable, multilingual, and accessible model for AI-based dermatological screening

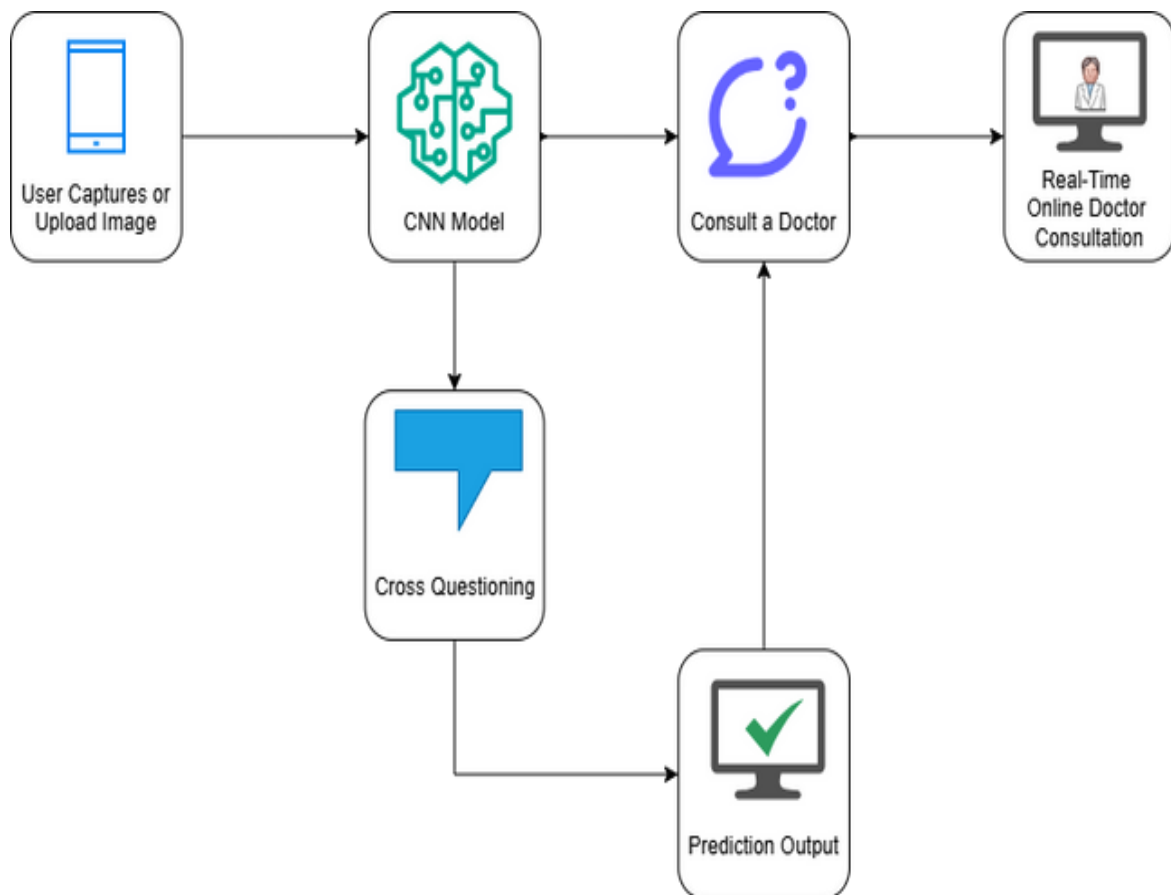
6. Methodology

This project will be developed as a cross-platform (desktop and mobile) application that helps users identify and comprehend eczema and fungal infections, two common skin disorders.

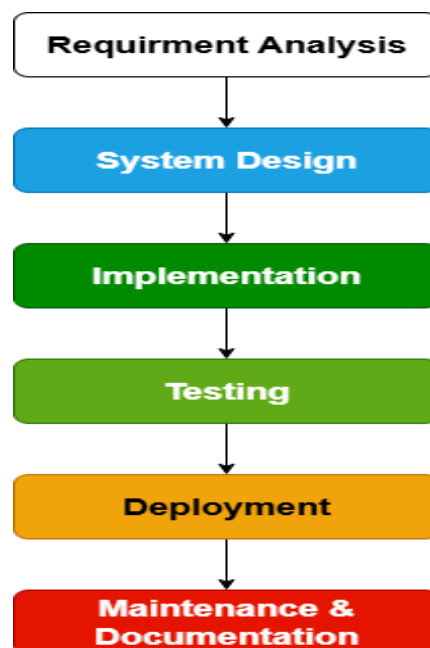
Three main elements make up the methodology: cross-questioning, image classification, and physician consultation. Convolutional neural networks (CNNs) will be used to first create a deep learning model by utilizing transfer learning to leverage pre-trained architectures like MobileNetV2 or EfficientNetB0. These models were picked because they strike a compromise between computational efficiency and accuracy. Curated dermatological image collections from sites like DermNet and ISIC will be used to refine the model, with an emphasis on eczema and fungal diseases. To enhance model performance and generalization, common image preprocessing methods such as scaling, normalization, and data augmentation (such as flipping and rotation) will be used.

The system will initiate a cross-questioning phase following the initial diagnostic based on the uploaded image. A Large Language Model (LLM) like GPT-3.5 or an open-source substitute will power this module. In order to obtain contextual information that might not be apparent in the picture, the LLM will pose specific symptom-based questions, such as "Is the area itchy?" or "Do you have a history of allergies?" In order to improve or confirm the initial prediction, the responses will be evaluated.

The user will be given the choice to "Consult a Doctor" once the final AI-generated diagnostic is complete. With the help of expert-verified responses, this function will mimic a consultation or link the user to a licensed medical practitioner via video call or chat. This increases user trust and guarantees medical oversight. Flutter (for mobile) and C# (for desktop) will be used in the development of the application. Data storage, LLM interaction, and AI model inference will be managed by backend services. The first iteration of the system will only support English and operate entirely online.



7. Workflow Diagram (Waterfall Model)



8. Feasibility Study

The system is thought to be possible within the academic timeline because of the project's well-defined scope, its narrow focus on two particular skin disorders (fungal infection and eczema), and its utilization of proven technology.

8.1 Dangers Associated- Limited Dataset Availability: There may be a lack or imbalance in some high-quality, annotated image data for eczema and fungal diseases. We will use publically accessible resources such as DermNet and ISIC to solve this, and we will use data augmentation approaches to increase the robustness of the model.

- Model Overfitting or Misclassification: There is a chance of making erroneous predictions because the classification of skin diseases is dependent on visual symptoms and picture quality. Cross-validation during model training, appropriate preprocessing, and transfer learning will all help to lessen this.- LLM Integration Challenges: Integrating a language model for symptom-based questions may result in irrelevant or imprecise responses. In order to solve this, we will limit open-ended interaction and create standardized, medically approved inquiry flows.- Doctor Consultation Integration: It could be difficult to integrate services or find a doctor in real time. If required, we will integrate with a third-party telehealth platform or use pre-written medical advice to mimic this feature.

8.2 Need for Resources- Hardware: An Android app(for testing mobile apps) and a mid-to-high-spec laptop or personal computer with GPU capabilities (for training models).- Software: cloud services for LLM (e.g., OpenAI API), Flutter for mobile user interfaces, Electron or C# for desktop user interfaces, and Python (TensorFlow/Keras).Human Resources: Basic assistance or input from a medical practitioner or student to confirm the cross-question design and medical reasoning.In conclusion, the project can be completed within the allotted academic time if it is properly planned and makes use of the tools and datasets that are accessible.

9. Solution Application Areas

In the areas of telemedicine and digital health, this project has substantial practical utility. In order to make skin disease screening more accessible, particularly in areas with restricted

access to dermatologists, it focuses on the fields of dermatological care and AI-assisted preliminary diagnosis.

The following domains are suitable for the successful implementation of the suggested system: Primary healthcare facilities for prompt patient triage and pre-diagnosis. Platforms for telemedicine that hope to provide skin assessments aided by AI. Applications for mobile health (mHealth) targeted at underserved or rural communities. Startups in the health technology sector are creating diagnostic solutions for non-specialist consumers. Medical and nursing students can learn about skin diseases from educational sites.

This approach can help close the gap between self-care and professional healthcare by enabling users to upload skin pictures, respond to questions about their symptoms, and receive AI-powered feedback before having the option to see a doctor. Particularly for treatable skin disorders like eczema and fungal infections, it can lower needless hospital stays, encourage early management, and raise patient awareness. Furthermore, this tool can be included into the offerings of telemedicine platforms and healthcare providers to improve patient interaction, expedite preliminary screening, and lessen burden. In conclusion, this project fills a glaring gap in digital healthcare and could benefit patients, caregivers, and healthcare providers in equal measure.

10. Tools/Technology

This project requires a combination of hardware and software tools to support image classification, natural language processing, user interface design, and overall system integration.

Hardware Requirements:

- - Personal Computer or Laptop (with at least 8GB RAM and GPU support for model training and testing)
- - Android Device (for testing the mobile version of the application)

Software / Technology Stack:

- 1. Machine Learning & Deep Learning:- Python 3.x- TensorFlow / Keras (for CNN-based image classification)- NumPy, Pandas, OpenCV (for data preprocessing and image handling)- Google Colab / Jupyter Notebook (for training and experimentation)

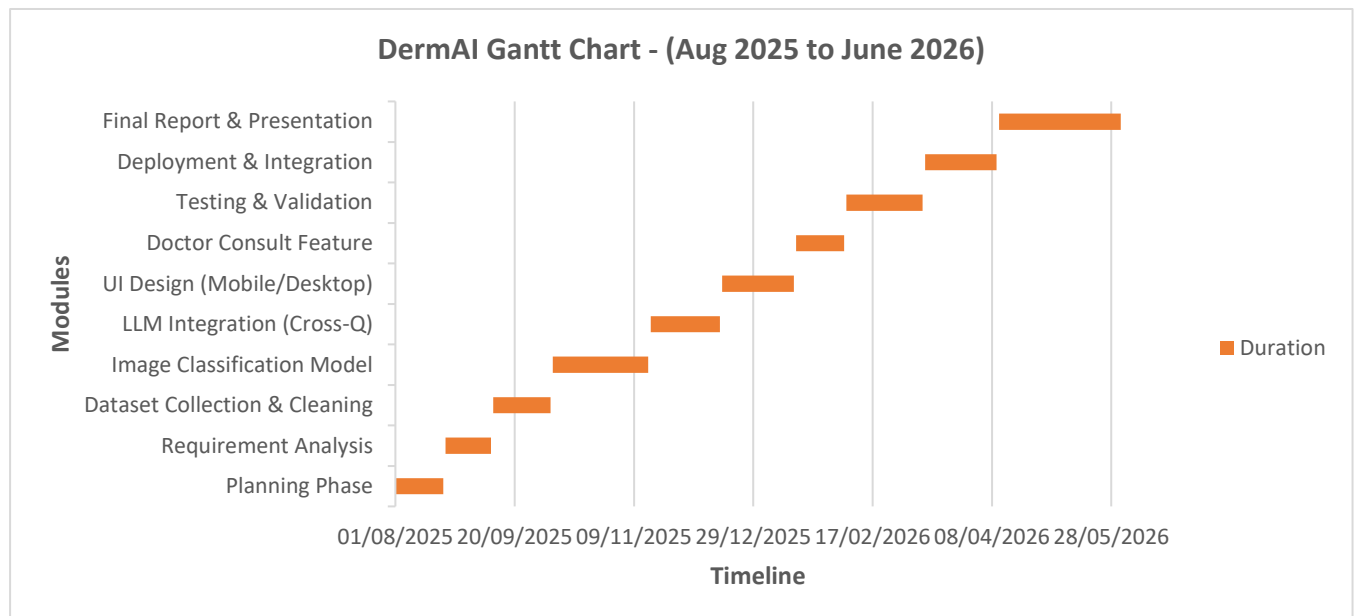
- . Natural Language Processing (NLP):- OpenAI GPT API / BERT (for LLM-powered symptom questioning)- NLTK / spaCy (for text handling and tokenization, if needed)
- . Mobile Application Development:- Flutter (cross-platform mobile app framework)- Dart (Flutter programming language)
- Desktop Application Development:- Electron.js (for building desktop apps using JavaScript and HTML)- or PyQt5 (if building with Python GUI framework)- Node.js (for Electron backend support)
- Cloud / Backend Services:- Firebase or Google Cloud Platform (for user authentication, image storage, and cloud inference)- OpenAI API or HuggingFace (for hosting language models)
- . Version Control & Collaboration:- Git & GitHub (for source code management and team collaboration)

These tools collectively support the end-to-end development of the proposed system, including training the AI models, designing intuitive user interfaces, managing user data securely, and integrating real-time doctor consultation capabilities.

11. Responsibilities of the Team Members

Task / Activity	Member 1 (Dataset & Model)	Member 2 (Mobile App & UI)	Member 3 (Testing and Report)	Supervisor
Dataset Collection & Cleaning	R	I	I	C
Model Selection & Training (CNN)	R	C	I	I
Model Conversion to TFLite	R	C	I	C
App UI Design & Navigation	I	R	C	I
Integration of AI Model into App	C	R	C	I
Chatbot or Doctor Module Integration	I	C	R	I
Text-to-Speech Module	I	C	R	C
Cross-Questioning Logic Design	C	C	R	C
Testing & Debugging	R	R	R	C
Documentation & Report Writing	R	R	R	I
Final App Demo Presentation	R	R	R	I

12. Gantt Chart



13. Project Budget

-  Internet & Firebase Services – PKR 10,000
Includes high-speed internet for testing & Firebase hosting for real-time data storage.
-  API Services & Online Tools – PKR 15,000
Covers services like Twilio (video integration), Jitsi server configuration, or other third-party APIs.
-  Audio/Video Tools & Content Creation – PKR 5,000
Microphone rental, basic recording setup for demo creation, and doctor session simulation.
-  Mobile Testing & Misc Devices – PKR 5,000
Covers cost of any accessories or device sharing required to test the app across different phones or Android versions.
-  Documentation, Printing, and Presentation Material – PKR 5,000
Includes report printing, binding, slide printing, and stationery for the final viva or university submission.

-  Final QA & Contingency – PKR 10,000
Buffer reserved for last-minute purchases, device testing failures, or hosting extensions.
-  **Total Estimated Cost: PKR 50,000**

14. Project Timeline & Milestones

Milestone 1: Planning Phase

- Establish team structure and responsibilities
- Select project title, objectives, and feasibility
- Identify tools, technologies, and dataset sources

Milestone 2: Requirement Analysis

- Conduct background research on dermatology and AI
- Define system requirements and expected features
- Document functional and non-functional specifications

Milestone 3: Dataset Collection & Cleaning

- Acquire HAM10000 / ISIC datasets
- Remove irrelevant data and preprocess images
- Label and organize the dataset for model input

Milestone 4: Image Classification Model

- Train and evaluate CNN (EfficientNet or MobileNetV2)
- Tune hyperparameters for optimal performance
- Convert trained model to TFLite for mobile integration

Milestone 5: LLM Integration (Cross-Questioning Module)

- Implement symptom-based dynamic questioning flow

- Integrate LLM (e.g., BERT or GPT) for follow-up prompts
- Validate contextual accuracy of user inputs

Milestone 6: UI Design (Mobile/Desktop)

- Design clean, responsive UI for both mobile and desktop
- Develop core app functionalities (image upload, results display)
- Connect UI to AI modules and backend services

Milestone 7: Doctor Consultation Feature

- Implement live video or asynchronous doctor consultation
- Use platforms like Firebase, Twilio, or third-party APIs
- Ensure consultation data is stored securely and ethically

Milestone 8: Testing & Validation

- Conduct system, integration, and user testing
- Evaluate performance, usability, and accessibility
- Fix bugs and refine user experience

Milestone 9: Deployment & Integration

- Deploy web and mobile apps with full functionality
- Monitor system logs, usage data, and user feedback
- Ensure all modules work in unison

Milestone 10: Final Report & Presentation

- Prepare final documentation and user manual
- Record a demo video explaining features and workflow
- Submit and present the complete project

15. References

- [1] Tschandl, Philipp, Cliff Rosendahl, and Harald Kittler. "The HAM10000 dataset, a large collection of multi-source dermoscopic images of common pigmented skin lesions." *Scientific Data*, 2018. <https://doi.org/10.1038/sdata.2018.161>. Last accessed May 13, 2025.
- [2] DermNet. "DermNet Image Library." No date. New Zealand Dermatological Society Incorporated. Last accessed May 13, 2025. <https://dermnetnz.org/image-library>
- [3] Tan, Mingxing, and Quoc V. Le. "EfficientNet: Rethinking model scaling for convolutional neural networks." *arXiv*, May 2019. Last accessed May 13, 2025. <https://arxiv.org/abs/1905.11946>
- [4] Singh, Rajeev. "AI Tools Set to Revolutionize Dermatology Diagnosis." *The Times of India*, January 12, 2023.

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